OFFICIAL APP ARCHITECTURE & OVERVIEW

FlowLedger (Expense Tracker)

A simple, beginner-friendly guide explaining the core features, underlying technologies, system architecture, and how the entire app works under the hood.

Application: FlowLedger v1.0

Framework: Flutter 3 (Dart)

Database: SQLite + Cloud Firestore

Platforms: Android, iOS, Web, Windows

01 What is FlowLedger?

FlowLedger is a calm, modern personal finance application built to help everyday people track where their money goes. Many expense apps are either cluttered with complicated financial jargon, aggressive advertisements, or require constant internet connections. FlowLedger was engineered around three golden principles:

- **Privacy & Offline-First:** Your money records live on your physical device first in a secure SQLite database. You can track expenses seamlessly on an airplane or without mobile data.
- **Effortless Visual Clarity:** Clean glassmorphic surfaces, automatic expense color-coding, high-contrast calendar views, and instant dark/light mode toggles.
- **Open Data Freedom:** You never get locked in. You can export complete, styled Microsoft Excel (`.xlsx`) spreadsheets anytime or import records from other apps (like *Expense Note* or custom Excel/CSV sheets).

02 Main Features in Simple Words

💸 1. Fast Transaction Logging

Quickly record an expense or income with amount, custom date, category (Food, Shopping, Bills, Transport, etc.), payment method (Card, UPI, Cash), and notes. Real-time balance calculations update instantly.

📊 2. Monthly Insights & Calendar

A visual calendar lets you tap any day of the month to see exactly what you spent that day. Interactive category breakdown bars show you what percentage of your budget went where.

📑 3. Excel & CSV Export/Import

Export your entire expense history into a styled `.xlsx` workbook with one tap and share it via WhatsApp, Email, or Drive. Easily import spreadsheets with automatic date and column recognition.

🌓 4. Synchronized Dark / Light Theme

Switch effortlessly between a deep space dark theme (`#090E1C`) and a crisp modern light theme. All app bars, calendars, cards, and text adapt with zero stutter.

🔄 5. Multi-Device Cloud Sync

Powered by Google Firebase Authentication and Cloud Firestore. If you sign in, your data backs up safely to the cloud so you can switch phones without losing a single rupee.

✨ 6. Figma Brand Splash & Logo

Designed faithfully to the brand identity: custom neon glowing "F" emblem squircle, corner ambient lighting, floating particles, and zero cold-start white screen flash on Android.

03 Technologies Used & Why We Picked Them

Every tool and library in FlowLedger was chosen for high performance, reliability, and ease of maintenance:

Technology	What It Does	Why It Was Chosen (Simple Reason)
Flutter & Dart	Cross-Platform UI Framework	Allows writing one single codebase that compiles into native high-performance apps for Android, iOS, Windows, and Web running at smooth 60/120 frames per second.
SQLite (sqflite)	Local On-Device Database	Stores all expenses directly inside the phone's memory. Extremely fast (microsecond lookups) and works 100% offline without needing internet or server servers.
Firebase Auth & Firestore	Cloud Accounts & Real-Time Sync	Provides secure email sign-in and real-time cloud database backup. When online, local changes automatically synchronize with the cloud.
Excel Package	Spreadsheet Generator & Parser	Generates real binary `.xlsx` files in memory and reads Excel rows without requiring Microsoft Office to be installed.
Share Plus (share_plus)	Native Android/iOS Sharing	Triggers the native operating system share sheet so you can directly send your Excel report to WhatsApp, Gmail, Telegram, or Google Drive.
File Picker	System File Dialogs	Allows the user to browse their device storage and pick `.xlsx` or `.csv` files when importing external data.
Custom Painter & Shaders	Glows & Gradients	Uses Flutter's GPU-accelerated graphics engine to draw smooth ambient lighting, glowing dots, and gradient text.

04 How the App Works (Under the Hood)

Here is what happens inside the application from the moment you tap an action on screen:

1. **User Enters an Expense**
   You tap "+" on the home screen, type ₹350 for Lunch, pick "Food", and tap Save.
2. **Immediate Local Persistence (SQLite)**
   The app instantly converts the entry into an `ExpenseRecord` and saves it to the SQLite database on your device. The home screen numbers update immediately with zero loading lag.
3. **Background Cloud Sync (Firebase)**
   If you are signed in and connected to Wi-Fi/data, a background service uploads a copy to your private Google Cloud Firestore collection. If offline, it waits safely until you reconnect.
4. **Live State Broadcast**
   All open screens (Home Dashboard, Insight Calendar, and Settings) receive a notification via Flutter's `Listenable / ChangeNotifier`, smoothly refreshing charts and balances.

> 💡 **Why Local-First Matters:** Even if you are in an underground metro or in a remote area with zero network reception, FlowLedger will never freeze, display spinning circles, or fail to save your expenses. Everything is saved locally first!

05 Key Challenges We Solved & How We Did It

During the development and refinement of FlowLedger, we addressed several technical challenges to ensure a world-class experience:

1. Synchronized Theme Switching

The Problem: When toggling dark mode, app bars in some screens changed color slightly faster than the body, causing visual stutter.
The Solution: We created a centralized `AppColor.revision` notifier and set `themeAnimationDuration: Duration.zero` on MaterialApp. Now, every layer updates in the exact same millisecond frame!

2. Calendar Text Contrast in Light Mode

The Problem: On the Insights page, calendar numbers became washed out or stayed dark-themed when in light mode.
The Solution: We wired calendar day tiles to listen dynamically to `AppColor.isDarkMode`, applying sharp high-contrast slate colors (`#1E293B`) in light mode and soft silver (`#E2E8F0`) in dark mode.

3. Smart Excel / Expense Note Import

The Problem: Different apps export dates in random formats (e.g. `2024-03-15`, `15/03/2024`, or raw Excel serial numbers like `45229`).
The Solution: We wrote a multi-strategy parser (`ExpenseNoteImporter`) that detects header names automatically, calculates Gregorian calendar dates from Excel serial numbers, and skips duplicate rows safely.

4. Figma Splash Screen & Android 12 Flash

The Problem: Android 12 launches with a default white circle flash before Flutter starts.
The Solution: We updated Android's native `LaunchTheme` to use `#090E1C`, and crafted a Flutter splash controller with curved ambient glows, emblem scaling, and a smooth cross-fade into Home.

06 Summary at a Glance

Aspect	Implementation Details
App Name	**FlowLedger** (registered across Android, iOS, Windows, Web, and Linux)
App Icon	Custom neon purple-to-teal squircle emblem built across all Android mipmap densities (mdpi to xxxhdpi)
Export Support	Native `.xlsx` spreadsheet with styled bold headers, formatted currency amounts, and share sheet
Import Support	Compatible with Microsoft Excel (`.xlsx`), CSV files, and direct migration from Expense Note
Theme Support	Seamless synchronized switching between Space Navy Dark (`#090E1C`) and Modern Light (`#F0F3F8`)

✨ **Conclusion:** FlowLedger combines the speed and privacy of an offline local database with the beauty of modern glassmorphic Flutter UI, making personal finance tracking enjoyable, fast, and stress-free.